EE4-18
EE9-AO6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2016

MSc and EEE PART IV: MEng and ACGI

RADIO FREQUENCY ELECTRONICS

Thursday, 12 May 10:00 am

Time allowed: 3:00 hours

Corrected copy

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : A.S. Holmes

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Special instructions for invigilators

This is a closed book examination.

Special instructions for students

None.

The Questions

1. An RF signal of 1 mW is input to a power amplifier with an output of 1 W. Design a suitable power amplifier to give the best overall performance using the following transistor stages.

$P_{OUT\|MAXLIN}$ [dBm]	P_{DC} [mW]	IP_3 [dBm]
25	600	40
30	2000	40
15	60	40

Table 1.1 Transistor Specifications.
(Note that all the transistors have perfect impedance matching)

[2]

a) From the design, calculate the following at each stage and the overall values for:

i) Power gain [1]

ii) Output power [2]

iii) Basic efficiency [2]

iv) PAE [2]

v) IP_3 [2]

vi) IMD_3 for two-tone power level given in (ii) [2]

vii) Dissipated power [2]

b) From first principles, prove that the 3rd order intermodulation log-power gain slope is three times that of the desired output log-power slope.

[2]

c) In linear operation, if the overall input power drops by 3 dB, what happens to the following:

i) Output power [1]

ii) I_3 power [1]

iii) IMD_3 [1]

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2. a) Describe, with the aid of a diagram, a branch-line coupler having a distributed-element implementation, indicating lengths and impedances. Indicate its main characteristics.

[5]

b) Replace the distributed-element implementation in 2(a) with an equivalent lumped-element version. How does the performance compare with that in 2(a)? With a coupler having an impedance of $Z_o = 50\ \Omega$, calculate the components values for an operating frequency of 1.8 GHz.

[5]

c) Replace the lumped-element implementation in 2(b) with an equivalent lumped-distributed version. With a coupler having an impedance of $Z_o = 50\ \Omega$ and line sections of $\phi = 45°$, calculate the components values for an operating frequency of 1.8 GHz.

[5]

d) Draw the topology of a balanced amplifier employing the coupler given in 2(a).

[5]

3.

a) Compare and contrast lumped-element impedance matching over the use of distributed-elements, in terms of frequency performance. How does this affect their role in DC biasing networks?

[4]

b) With lumped-element impedance matching, what terminating impedance conditions are best suited for L-match, π-match and T-match networks.

[4]

c) A 2 GHz narrow-band amplifier has an output impedance of $(5 - j7)\ \Omega$ and must be matched to a system impedance of 50 Ω. Design simple matching circuits to achieve maximum power transfer:

i) With the use of one lumped-element component and one distributed-element component

[4]

ii) With the use of two lumped-element components

[4]

d) With monolithic technology, discuss the difficulties implementing lumped-element and distributed-element components. How can micromachining technologies help to overcome some of the problems with monolithic implementations? How does this affect their role in filter networks?

[4]

4.

a) Explain why filters having sharp frequency roll-off characteristics require large components to achieve low insertion losses.

[4]

b) Explain why impedance and admittance inverters are required for realising practical narrow bandwidth filters. In addition, with the use of simple block diagrams, explain how these inverters work.

[6]

c) Redesign the 1.8 GHz resonator topology shown in Figure 4.1, by employing all capacitor admittance inverters, so that the series tuned circuit can be replaced with a shunt parallel tuned circuit.

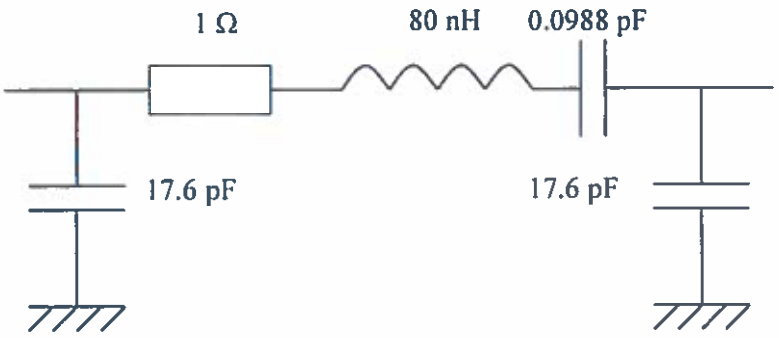


Figure 4.1: Lumped-element resonator.

[8]

d) What effect on the insertion phase does the all capacitor admittance inverter have?

[2]

5.

(a) Given a square wave signal generator, with a clock frequency that can vary between DC and 1000 MHz, propose how this could be used to generate a sinusoidal signal at a frequency of 1800 MHz.

[4]

(b) Design a lumped-element *L-C* high-pass filter to meet the following specifications:

Pass band attenuation ripple	0.1 dB
-3 dB cut-off frequency:	1500 MHz
Stop band frequency:	600 MHz
Stop band attenuation:	> 70 dB
Source impedance, Zs:	50 Ω
Load impedance, Z_L:	50 Ω

[10]

(c) Determine the worst-case levels of return losses within both the pass band and the stop band for the filter in 5(b). How could the stop band return loss adversely affect the implementation of 5(a) and suggest a suitable topology for overcoming this problem?

[6]

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6.

a) With the aid of a diagram, describe the S-parameter representation of a linear two-port circuit, stating the precise definitions of all parameters and the main power specifications.

[5]

b) State which RF components best described the following S-parameter matrices and calculate any relevant power specifications:

(i) $$[S]=\begin{pmatrix} 0 & e^{-j720} \\ e^{-j720} & 0 \end{pmatrix} \qquad (6.1)$$

(ii) $$[S]=\begin{pmatrix} 0 & 0.07e^{-j30} \\ e^{-j60} & 0 \end{pmatrix} \qquad (6.2)$$

(iii) $$[S]=\begin{pmatrix} 0.1e^{-j30} & 0.3e^{-j80} \\ 9.7e^{-j80} & 0.15e^{-j60} \end{pmatrix} \qquad (6.2)$$

[5]

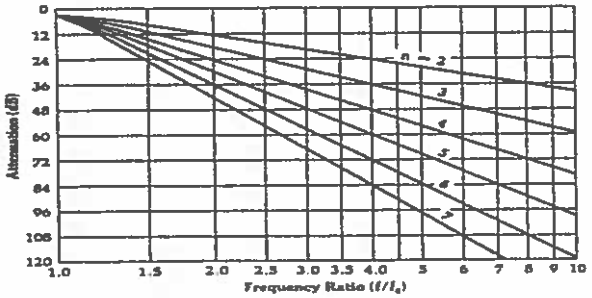
c) Derive an algebraic expression for the overall Γ_{in}, of a linear two-port network that is terminated at its output port with a one-port network represented by Γ_L.

[6]

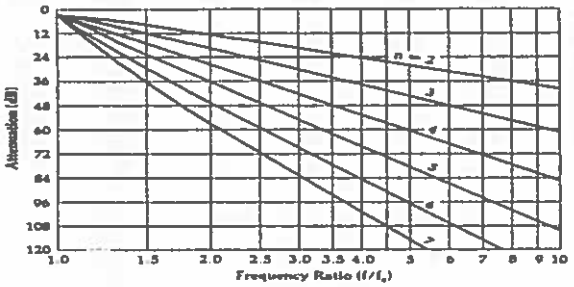
d) Referring to the result in 6(c), state the condition for stability for the overall one-port network, for any value of generator source impedance. If the two-port network in 6(b)(ii) is terminated with a load impedance having $\Gamma_L = 0.5$, determine if the overall one-port network is stable.

[4]

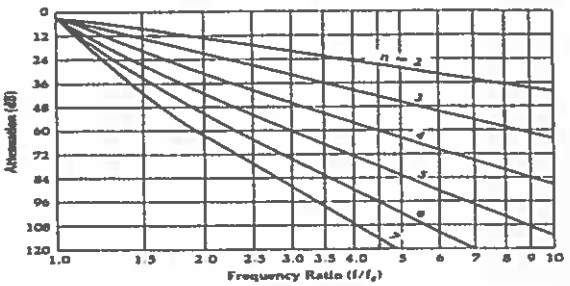
Standard Filter Curves and Tables



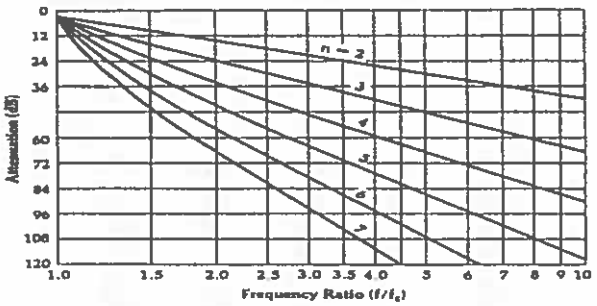
Attenuation characteristics for Butterworth filters.



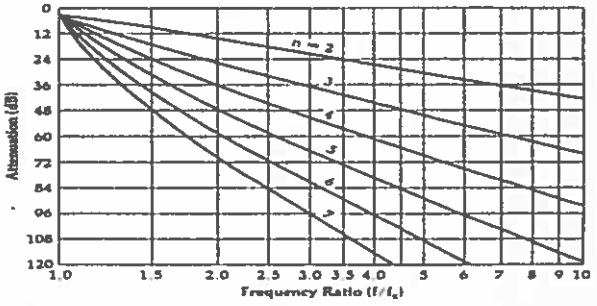
Attenuation characteristics for a Chebyshev filter with 0.01-dB ripple.



Attenuation characteristics for a Chebyshev filter with 0.1-dB ripple.

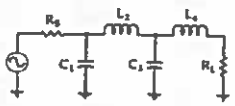


Attenuation characteristics for a Chebyshev filter with 0.5-dB ripple.



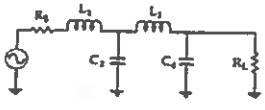
Attenuation characteristics for a Chebyshev filter with 1-dB ripple.

Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

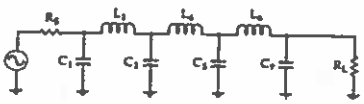


R_s/R_L	C_1	L_2	C_3	L_4
3.000	0.572	3.132		
4.000	0.365	4.000		
8.000	0.157	9.658		
∞	1.213	1.100		
1.000	2.216	1.088	2.216	
0.500	4.431	0.817	2.216	
0.333	6.647	0.726	2.216	
0.250	8.862	0.680	2.216	
0.125	17.725	0.612	2.216	
∞	1.652	1.460	1.108	
3.000	0.653	4.411	0.814	2.535
4.000	0.452	7.053	0.612	2.848
8.000	0.209	17.164	0.428	3.281
∞	1.350	2.010	1.466	1.106

R_L/R_s	L_1	C_2	L_3	C_4

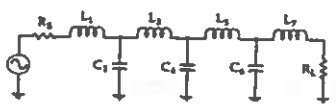


Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple



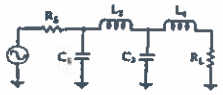
n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	2.807	1.128	3.103	1.128	2.807		
	0.500	4.414	0.565	4.653	1.128	2.807		
	0.333	6.622	0.376	6.205	1.128	2.807		
	0.250	8.829	0.282	7.756	1.128	2.807		
	0.125	17.657	0.141	13.961	1.128	2.807		
	∞	1.721	1.645	2.061	1.493	1.103		
6	3.000	0.679	3.673	0.771	4.711	0.969	2.406	
	4.000	0.481	5.044	0.476	7.361	0.849	2.582	
	8.000	0.227	12.310	0.198	16.740	0.726	2.800	
	∞	1.378	2.097	1.590	2.074	1.494	1.102	
7	1.000	2.204	1.131	3.147	1.194	3.147	1.131	2.204
	0.500	4.406	0.566	6.293	0.995	3.147	1.131	2.204
	0.333	6.613	0.377	9.441	0.798	3.147	1.131	2.204
	0.250	8.815	0.283	12.586	0.747	3.147	1.131	2.204
	0.125	17.631	0.141	25.173	0.671	3.147	1.131	2.204
	∞	1.741	1.677	2.155	1.703	2.079	1.494	1.102

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7



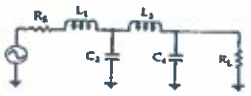
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Butterworth Low-Pass Prototype Element Values

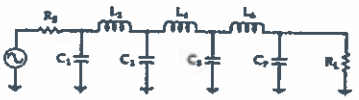


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.111	1.035	1.835		
	1.250	0.849	2.121		
	1.429	0.697	2.439		
	1.667	0.566	2.828		
	2.000	0.448	3.346		
	2.500	0.342	4.095		
	3.333	0.245	5.313		
	5.000	0.156	7.707		
	10.000	0.074	14.814		
	∞	1.414	0.707		
3	0.900	0.808	1.633	1.599	
	0.800	0.844	1.364	1.926	
	0.700	0.915	1.163	2.277	
	0.600	1.023	0.965	2.702	
	0.500	1.181	0.779	3.261	
	0.400	1.425	0.604	4.064	
	0.300	1.838	0.440	5.363	
	0.200	2.669	0.284	7.910	
	0.100	5.187	0.135	15.455	
	∞	1.500	1.333	0.500	
4	1.111	0.466	1.592	1.744	1.469
	1.250	0.388	1.695	1.511	1.811
	1.429	0.325	1.662	1.391	2.175
	1.667	0.269	2.103	1.062	2.613
	2.000	0.218	2.432	0.863	3.187
	2.500	0.169	2.986	0.691	4.009
	3.333	0.124	3.883	0.507	5.338
	5.000	0.080	5.684	0.331	7.940
	10.000	0.039	11.064	0.162	15.642
	∞	1.531	1.577	1.082	0.383

n	R_L/R_s	L_1	C_2	L_3	C_4

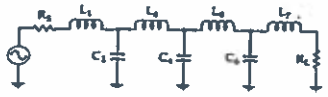


Butterworth Low-Pass Prototype Element Values

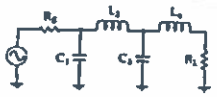


n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	0.900	0.442	1.027	1.910	1.756	1.389		
	0.800	0.470	0.866	2.061	1.544	1.738		
	0.700	0.517	0.731	2.285	1.333	2.108		
	0.600	0.586	0.609	2.600	1.126	2.552		
	0.500	0.686	0.496	3.051	0.924	3.133		
	0.400	0.838	0.388	3.736	0.727	3.985		
	0.300	1.094	0.285	4.884	0.537	5.307		
	0.200	1.608	0.186	7.185	0.352	7.935		
	0.100	3.519	0.091	14.095	0.173	15.710		
	∞	1.545	1.894	1.382	0.894	0.309		
6	1.111	0.289	1.040	1.322	2.054	1.744	1.335	
	1.250	0.245	1.116	1.126	2.239	1.580	1.688	
	1.429	0.207	1.236	0.957	2.499	1.346	2.063	
	1.667	0.173	1.407	0.801	2.858	1.143	2.500	
	2.000	0.141	1.653	0.654	3.369	0.942	3.094	
	2.500	0.111	2.028	0.514	4.141	0.745	3.931	
	3.333	0.082	2.656	0.379	5.433	0.552	5.280	
	5.000	0.054	3.917	0.248	8.020	0.363	7.922	
	10.000	0.026	7.705	0.122	15.786	0.179	15.738	
	∞	1.553	1.759	1.553	1.202	0.758	0.259	
7	0.900	0.299	0.711	1.404	1.489	2.125	1.787	1.396
	0.800	0.322	0.606	1.517	1.378	2.334	1.546	1.652
	0.700	0.357	0.515	1.686	1.091	2.618	1.350	2.028
	0.600	0.406	0.433	1.928	0.917	3.005	1.150	2.477
	0.500	0.480	0.354	2.273	0.751	3.553	0.951	3.064
	0.400	0.590	0.278	2.795	0.592	4.380	0.754	3.904
	0.300	0.775	0.206	3.671	0.437	5.761	0.560	5.258
	0.200	1.145	0.135	5.427	0.287	8.526	0.369	7.908
	0.100	2.257	0.067	10.700	0.143	16.832	0.183	15.748
	∞	1.536	1.799	1.659	1.397	1.055	0.656	0.223

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7

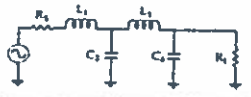


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

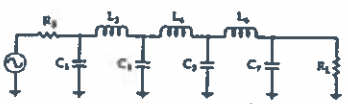


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.101	1.347	1.483		
	1.111	1.347	1.505		
	1.250	0.943	1.997		
	1.429	0.750	2.344		
	1.667	0.600	2.750		
	2.000	0.479	3.277		
	2.500	0.383	4.033		
	3.333	0.289	5.253		
	5.000	0.184	7.650		
	10.000	0.078	14.749		
	∞	1.412	0.742		
3	1.000	1.181	1.821	1.181	
	0.900	1.092	1.660	1.460	
	0.800	1.097	1.443	1.806	
	0.700	1.160	1.228	2.165	
	0.600	1.274	1.024	2.598	
	0.500	1.452	0.829	3.164	
	0.400	1.734	0.645	3.974	
	0.300	2.219	0.470	5.250	
	0.200	3.193	0.305	7.834	
	0.100	6.141	0.148	15.290	
	∞	1.501	1.433	0.591	
4	1.100	0.950	1.938	1.761	1.046
	1.111	0.854	1.946	1.744	1.165
	1.250	0.618	2.075	1.542	1.617
	1.429	0.493	2.279	1.334	2.008
	1.667	0.368	2.571	1.128	2.461
	2.000	0.316	2.964	0.926	3.045
	2.500	0.242	3.641	0.729	3.875
	3.333	0.174	4.727	0.538	5.209
	5.000	0.112	6.910	0.352	7.813
	10.000	0.054	13.469	0.173	15.510
	∞	1.529	1.694	1.312	0.523

n	R_L/R_s	L_1	C_2	L_3	C_4

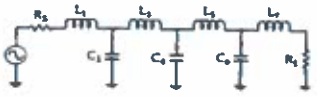


Chebyshev Low-Pass Element Values for 0.01-dB Ripple



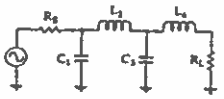
n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	0.977	1.685	2.037	1.685	0.977		
	0.900	0.880	1.456	2.174	1.641	1.274		
	0.800	0.877	1.235	2.379	1.499	1.607		
	0.700	0.926	1.040	2.658	1.323	1.977		
	0.600	1.019	0.863	3.041	1.135	2.494		
	0.500	1.166	0.699	3.584	0.943	3.099		
	0.400	1.398	0.544	4.403	0.749	3.845		
	0.300	1.797	0.398	5.772	0.557	5.193		
	0.200	2.604	0.259	8.514	0.368	7.826		
	0.100	5.041	0.127	16.741	0.182	15.813		
	∞	1.547	1.795	1.845	1.237	0.488		
6	1.101	0.851	1.796	1.841	2.037	1.631	0.937	
	1.111	0.760	1.782	1.775	2.094	1.636	1.053	
	1.250	0.545	1.864	1.489	2.403	1.507	1.504	
	1.429	0.436	2.035	1.266	2.725	1.332	1.899	
	1.667	0.351	2.298	1.061	3.167	1.145	2.357	
	2.000	0.279	2.678	0.867	3.766	0.954	2.948	
	2.500	0.214	3.261	0.682	4.667	0.761	3.790	
	3.333	0.155	4.245	0.503	6.163	0.566	5.143	
	5.000	0.100	6.233	0.330	9.151	0.376	7.785	
	10.000	0.048	12.171	0.162	16.105	0.187	15.595	
	∞	1.551	1.847	1.790	1.596	1.190	0.469	
7	1.000	0.913	1.595	2.002	1.870	2.002	1.595	0.913
	0.900	0.816	1.362	2.089	1.722	2.202	1.581	1.206
	0.800	0.811	1.150	2.262	1.523	2.485	1.464	1.538
	0.700	0.857	0.967	2.516	1.323	2.802	1.307	1.910
	0.600	0.943	0.803	2.872	1.124	3.250	1.131	2.369
	0.500	1.080	0.650	3.382	0.928	3.875	0.947	2.948
	0.400	1.297	0.507	4.156	0.735	4.812	0.758	3.790
	0.300	1.669	0.372	5.454	0.546	6.370	0.568	5.148
	0.200	2.242	0.242	8.057	0.360	9.484	0.378	7.808
	0.100	4.701	0.119	15.572	0.178	18.816	0.188	15.852
	∞	1.559	1.867	1.866	1.765	1.563	1.161	0.456

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7



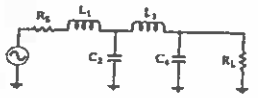
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Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

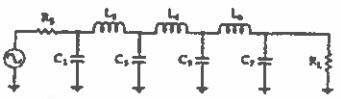


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.333	1.209	1.638		
	1.429	0.977	1.982		
	1.667	0.733	2.480		
	2.000	0.560	3.054		
	2.500	0.417	3.827		
	3.333	0.293	5.050		
	5.000	0.184	7.426		
	10.000	0.087	14.433		
	∞	1.301	0.819		
3	1.000	1.433	1.594	1.433	
	0.900	1.426	1.494	1.622	
	0.800	1.451	1.356	1.871	
	0.700	1.521	1.193	2.190	
	0.600	1.648	1.017	2.603	
	0.500	1.853	0.838	3.159	
	0.400	2.186	0.660	3.968	
	0.300	2.763	0.486	5.279	
	0.200	3.842	0.317	7.850	
	0.100	7.512	0.155	15.466	
	∞	1.513	1.510	0.716	
4	1.333	0.992	2.148	1.585	1.341
	1.429	0.779	2.348	1.429	1.700
	1.667	0.576	2.730	1.185	2.343
	2.000	0.440	3.227	0.967	3.256
	2.500	0.329	3.961	0.760	4.398
	3.333	0.233	5.178	0.560	6.030
	5.000	0.148	7.607	0.367	9.814
	10.000	0.070	14.887	0.180	19.230
	∞	1.511	1.768	1.455	0.873

n	R_L/R_s	L_1	C_2	L_3	C_4

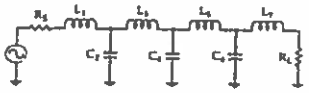


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

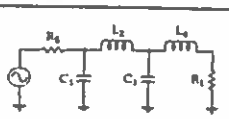


n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.301	1.536	2.941	1.536	1.301		
	0.900	1.285	1.433	2.360	1.486	1.488		
	0.800	1.300	1.283	2.582	1.382	1.736		
	0.700	1.358	1.117	2.866	1.244	2.082		
	0.600	1.470	0.947	3.269	1.065	2.584		
	0.500	1.654	0.778	3.845	0.913	3.355		
	0.400	1.954	0.613	4.780	0.733	3.886		
	0.300	2.477	0.451	6.196	0.580	5.337		
	0.200	3.546	0.295	9.127	0.366	7.896		
	0.100	6.787	0.115	17.957	0.182	15.745		
	∞	1.561	1.807	1.766	1.417	0.651		
6	1.355	0.948	2.080	1.659	2.247	1.534	1.277	
	1.429	0.735	2.249	1.454	2.544	1.405	1.629	
	1.667	0.543	2.600	1.183	3.064	1.185	2.174	
	2.000	0.414	3.068	0.958	3.712	0.979	2.704	
	2.500	0.310	3.785	0.749	4.651	0.778	3.645	
	3.333	0.220	4.927	0.551	6.195	0.580	4.996	
	5.000	0.139	7.250	0.361	9.261	0.384	7.618	
	10.000	0.067	14.220	0.178	18.437	0.190	15.350	
	∞	1.534	1.884	1.831	1.749	1.394	0.838	
7	1.000	1.262	1.520	2.829	1.650	2.829	1.520	1.262
	0.900	1.248	1.398	2.581	1.578	2.397	1.459	1.447
	0.800	1.255	1.245	2.548	1.443	2.824	1.303	1.667
	0.700	1.310	1.083	2.819	1.253	2.942	1.133	2.021
	0.600	1.417	0.917	3.205	1.009	3.364	1.061	2.444
	0.500	1.595	0.753	3.764	0.928	4.015	0.914	3.018
	0.400	1.885	0.593	4.818	0.742	4.970	0.738	3.855
	0.300	2.392	0.437	6.654	0.556	6.569	0.557	5.217
	0.200	3.428	0.286	9.827	0.369	9.770	0.372	7.800
	0.100	6.570	0.141	17.803	0.184	19.376	0.186	15.813
	∞	1.575	1.858	1.921	1.827	1.734	1.379	0.631

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7

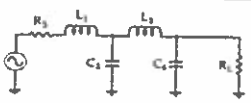


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple

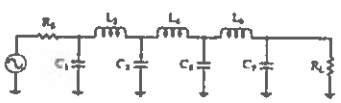


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.984	0.983	1.950		
	2.000	0.909	2.103		
	2.500	0.564	3.165		
	3.333	0.375	4.411		
	5.000	0.228	6.700		
	10.000	0.105	13.322		
	∞	1.307	0.975		
3	1.000	1.864	1.280	1.834	
	0.900	1.918	1.209	2.026	
	0.800	1.997	1.120	2.237	
	0.700	2.114	1.015	2.517	
	0.500	2.557	0.759	3.436	
	0.400	2.985	0.615	4.342	
	0.300	3.729	0.463	5.376	
	0.200	5.254	0.309	8.225	
	0.100	9.890	0.153	16.116	
	∞	1.572	1.518	0.932	
4	1.984	0.920	2.586	1.304	1.828
	2.000	0.845	2.720	1.238	1.985
	2.500	0.516	3.766	0.869	3.121
	3.333	0.344	5.120	0.621	4.480
	5.000	0.210	7.706	0.400	6.987
	10.000	0.098	15.358	0.194	14.362
	∞	1.436	1.889	1.521	0.913

n	R_L/R_s	L_1	C_2	L_3	C_4

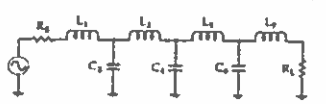


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple



n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.807	1.303	2.891	1.303	1.807		
	0.900	1.854	1.222	2.849	1.238	1.970		
	0.800	1.928	1.126	3.060	1.157	2.185		
	0.700	2.035	1.015	3.353	1.058	2.470		
	0.600	2.200	0.890	3.765	0.942	2.861		
	0.500	2.457	0.754	4.367	0.810	3.414		
	0.400	2.870	0.609	5.296	0.664	4.245		
	0.300	3.588	0.459	6.871	0.508	5.625		
	0.200	5.064	0.306	10.054	0.343	8.367		
	0.100	9.556	0.153	19.647	0.173	16.574		
	∞	1.630	1.740	1.922	1.514	0.903		
6	1.984	0.905	2.577	1.366	2.713	1.299	1.796	
	2.000	0.830	2.704	1.291	2.872	1.237	1.956	
	2.500	0.506	3.722	0.890	4.109	0.881	3.103	
	3.333	0.337	5.055	0.632	5.999	0.635	4.481	
	5.000	0.206	7.615	0.400	8.732	0.418	7.031	
	10.000	0.096	15.186	0.197	17.681	0.202	14.433	
7	1.000	1.790	1.296	2.718	1.385	2.718	1.296	1.790
	0.900	1.835	1.215	2.849	1.306	2.883	1.234	1.953
	0.800	1.905	1.118	3.076	1.215	3.107	1.155	2.168
	0.700	2.011	1.007	3.364	1.105	3.418	1.058	2.455
	0.600	2.174	0.882	3.772	0.979	3.852	0.944	2.848
	0.500	2.428	0.747	4.370	0.838	4.289	0.814	3.405
	0.400	2.835	0.604	5.296	0.685	5.470	0.669	4.243
	0.300	3.546	0.455	6.867	0.522	7.134	0.513	5.535
	0.200	5.007	0.303	10.049	0.352	10.498	0.348	8.404
	0.100	9.456	0.151	19.649	0.178	20.631	0.176	16.665
	∞	1.646	1.777	2.031	1.789	1.924	1.503	0.895

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7



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